Run report: dreg closure (4,6), run 20261018T1134Z

Study `research/dreg_closure_m4l6_20261018/`, host RunPod `wfh2xovuq5aahv` (25 vCPU, cgroup 200 GB, cap A = 184 GB), commit `9d4541e2`, draws commit `fc1eea79` (sha256 `d472512a…`). Observations only; the §4 reading is in `RESULTS.md`.

Runs

run	cell	draw	to	outcome	wall	peak RSS
measure	A1 (4; 15, 6)	42 (unsat)	4	exit 0, closure refuted at 4	24:27	25.8 GB
measure	A1	214 (unsat)	4	exit 0, closure refuted at 4	23:36	25.7 GB
deg5	C1 (3; 19, 7)	0 (unsat)	5	failed: allocation of 94,976 bytes failed, SIGABRT, no record	17:28	169.3 GB
control	A1	0 (sat, 1,226 solutions)	4	exit 0, not refuted	34:36	21.4 GB

Wall and RSS from GNU `time` (measured). The two A1 draws ran one after the other (§8 paired A1 with A2; A2 is deferred by §10).

Recorded scans

draw	deg	columns	XL rank	closure rank	XL ref.	closure ref.	products	word ops ($\times 10^6$)	secs
42	3	9,022	854	2,748	no	no	5,539	24.1	0.197
42	4	341,326	23,757	1	no	**yes**	1,181,122	2,873,337	1,464.66
214	3	9,022	854	2,748	no	no	5,542	24.3	0.214
214	4	341,056	23,757	1	no	**yes**	1,178,721	2,455,694	1,413.40
0 ctrl	3	11,662	854	2,748	no	no	5,643	26.9	0.217
0 ctrl	4	342,537	23,757	214,391	no	no	1,094,741	2,712,063	2,072.76

Closure rank 1 is the engine’s refutation signal; exact word-op counts are in `README.md`. Sample: 2 unsatisfiable draws (all found in 251 at S = −9) and 1 satisfiable control; C1 has no completed run.

Against the projections

A1 peak RSS 25.8 GB against 44 GB projected (0.59×). C1 at degree 5 ran out of memory at 169 GB under the 184 GB cap, against a 217 GB projection: capped on this host. Unsatisfiable-draw rate at S = −9: 2/251.

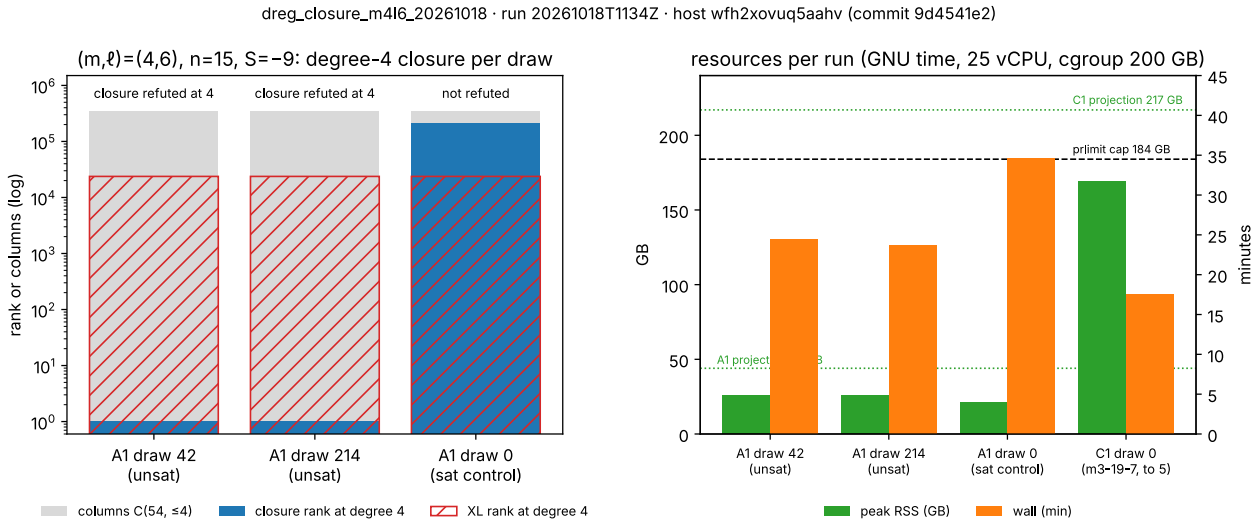


Figure 1: Run 20261018T1134Z, host `wfh2xovuq5aahv`. Left: columns, XL rank and closure rank at degree 4 per draw (log scale). Right: peak RSS and wall per run with the §2 projections and the 184 GB cap. Recorded values only (`plot.py`).

Missing

A2, B1, B2, C2 not run (deferred). C1: no completed run. A1: only two unsatisfiable draws exist. Raw records in `runs/`; generation log `draws/m4-15-6.log`.